



NAME:
HABIB ID:
SECTION:
INSTRUCTOR:

LINEAR ALGEBRA

SPRING 2026

QUIZ 15 (30th April 2026)

Max Marks: 10

Time: 12 minutes

a. Define a Linear Transformation.

[2]

b. Consider the mapping $F\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x + y + z \end{bmatrix}$.

What is the domain of F , and what is the image space of F ?

[2]

c. Using your definition in part (a), check whether F is a linear transformation or not.

[6]



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a. Define a **Linear Transformation**.

[2]

b. Consider the mapping $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x + y + z \end{bmatrix}$.

What is the domain of T , and what is the image space of T ?

[2]

c. Using your definition in part (a), check whether T is a linear transformation or not.

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SOLUTION

Part a

IF $f: V \rightarrow W$ IS A FUNCTION FROM THE VECTOR SPACE V INTO THE VECTOR SPACE W , THEN f IS CALLED A LINEAR TRANSFORMATION IF

$$(a) \quad f(\underline{u} + \underline{v}) = f(\underline{u}) + f(\underline{v})$$

FOR ALL $\underline{u}, \underline{v} \in V$

$$(b) \quad f(k\underline{u}) = k f(\underline{u}) \text{ FOR ALL } \underline{u} \in V \text{ AND ALL SCALARS } k.$$

Part b

Domain is $\mathbb{R}^3$

Image space is $\mathbb{R}^2$

Part c

check the 2 properties for linear transformations

$$F: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$(1) \text{ Let } \underline{u} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \text{ and } \underline{v} = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} \in \mathbb{R}^3 \Rightarrow \underline{u} + \underline{v} \in \mathbb{R}^3$$

$$\begin{aligned} F(\underline{u} + \underline{v}) &= \begin{bmatrix} x_1 + x_2 \\ x_1 + x_2 + y_1 + y_2 + z_1 + z_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ x_1 + y_1 + z_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ x_2 + y_2 + z_2 \end{bmatrix} = F(\underline{u}) + F(\underline{v}) \end{aligned}$$

$$\begin{aligned} (2) \quad F(k\underline{u}) &= F \begin{bmatrix} kx_1 \\ ky_1 \\ kz_1 \end{bmatrix} = \begin{bmatrix} kx_1 \\ kx_1 + ky_1 + kz_1 \end{bmatrix} \\ &= \begin{bmatrix} kx_1 \\ k(x_1 + y_1 + z_1) \end{bmatrix} = k \begin{bmatrix} x_1 \\ x_1 + y_1 + z_1 \end{bmatrix} \\ &= k F(\underline{u}) \end{aligned}$$

$\therefore F$ is a linear mapping / transformation